

Strategic Marketing Storyboard

Network Centrality Analysis for Content Strategy

Data-Driven Insights for Multi-Genre Positioning

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Role: Digital Marketing Specialist

Project: Movie Network Analysis

Focus: Genre-Based Audience Targeting

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Executive Summary

Ana Sofía Mendoza leverages network centrality analysis to uncover strategic insights about content positioning and audience reach in the entertainment industry. By analyzing 383 movies through four centrality metrics (degree, closeness, betweenness, and PageRank), this study reveals that genre diversity—not popularity—drives network influence. The findings provide actionable strategies for content marketing, multi-platform positioning, and cross-genre audience engagement.

Key Discovery

Movies with highest network centrality achieve influence through genre hybridization, connecting diverse audience segments. This insight transforms how Ana Sofía approaches campaign strategy: successful content marketing requires understanding structural position within genre ecosystems, not just measuring raw popularity metrics.

1 Project Context

The Challenge

In digital marketing for entertainment content, traditional metrics like popularity and ratings often dominate strategy decisions. However, these metrics fail to capture how content actually connects audiences across different interest communities. Ana Sofía needed to understand:

- Which content serves as "bridge" between different audience segments?
- How does genre mixing affect potential audience reach?
- Can network structure predict content influence better than popularity?
- What positioning strategies maximize cross-audience exposure?

The Approach

Using network analysis techniques on 383 movies from popular and top-rated datasets, Ana Sofía built a genre-based connection network where:

- **Nodes:** Individual movies with attributes (rating, popularity, vote count)
- **Edges:** Genre overlap connections weighted by number of shared genres
- **Analysis:** Four centrality metrics to measure different types of influence

2 Network Analysis Framework

Centrality Metrics Explained

Four Dimensions of Influence

Degree Centrality (0.456 avg): Direct connections—how many other movies share genres

Closeness Centrality (0.652 avg): Accessibility—average distance to all other content

Betweenness Centrality (0.001 avg): Bridge position—connects otherwise separate communities

PageRank (0.003 avg): Influence quality—importance of connected nodes

Network Structure Insights

Ultra-Dense Network Discovered:

Diameter: 3 steps — Avg. Path Length: 1.55 — Clustering: 0.774

Avg. Connections per Movie: 174 out of 382 possible

Marketing Implication: Any content is maximum 3 genre-steps away from any other content. Cross-promotion opportunities are everywhere, but standing out requires strategic differentiation.

3 Storyboard: Ana Sofía's Discovery Journey

Scene 1: The Popularity Paradox

Visual: Dashboard comparing popularity vs. network centrality

Data Point: "Steven Universe: The Movie" has highest centrality (1.0) but lowest popularity (2.9)

Ana Sofía's Insight: "Traditional metrics are misleading. The most connected content isn't the most popular—it's the most genre-diverse. This changes everything about how we position campaigns."

Scene 2: The Genre Hub Discovery

Visual: Network graph showing Drama (Genre 18) as central node with 171 connections

Data Point: Drama + Thriller (101) + Action (96) = Universal connector pattern

Strategic Realization: "Drama isn't just a genre—it's the backbone that connects all audience segments. Any content with dramatic elements automatically gains wider network reach. We need to emphasize dramatic storytelling even in action or comedy campaigns."

Scene 3: The Bridge Opportunity

Visual: Correlation heatmap showing degree-closeness correlation at 0.990

Analysis: High correlation means most central nodes are "hubs," not "bridges"

Marketing Pivot: "Django Unchained, Frankenstein, and The Substance have high betweenness—they uniquely bridge communities. These are perfect for cross-promotional campaigns targeting multiple audience segments simultaneously."

Scene 4: The ROI Disconnect

Visual: Scatter plots showing centrality vs. popularity correlation at -0.064

Critical Finding: Network position has zero correlation with commercial performance

Strategy Formulation: "Centrality predicts influence potential, not sales. For awareness campaigns, target high-centrality content. For conversion campaigns, prioritize high-popularity. Different objectives need different network positions."

4 Key Findings & Marketing Implications

Finding 1: Genre Diversity = Network Power

Data: Top 5 central movies average 0.82 composite score with multi-genre profiles

Example: "The Dark Knight" (Drama + Action + Crime) achieves 0.75 centrality

Marketing Strategy:

- Position content by emphasizing genre hybridization in messaging
- Create separate campaigns for each genre community the content bridges
- Use multi-genre tags in paid media to capture diverse search intent
- A/B test creative that highlights different genre elements for different audiences

Finding 2: Drama as Universal Language

Data: Genre 18 (Drama) appears in 171/383 movies (45% of network)

Correlation: Drama presence correlates with higher degree centrality

Marketing Strategy:

- Always include "emotional stakes" framing in content marketing
- Lead with dramatic narrative in video ads, even for action/comedy content
- Target drama-loving audiences first, then expand to niche genres
- Use dramatic testimonials and user stories in social proof campaigns

Finding 3: The Centrality-Popularity Gap

Data: Centrality vs Popularity correlation = -0.064 (essentially zero)

Example: "Steven Universe" (centrality: 1.0, popularity: 2.9) vs typical patterns

Marketing Strategy:

- **Awareness Stage:** Promote high-centrality content to seed multiple communities
- **Consideration Stage:** Showcase bridges (high betweenness) for cross-sell
- **Conversion Stage:** Focus budget on high-popularity content with proven appeal
- **Retention Stage:** Use centrality to recommend related content across genres

Finding 4: Small-World Structure

Data: Diameter = 3, Avg Path = 1.55, Clustering = 0.774

Implication: Every movie is 1-2 steps from any other movie

Marketing Strategy:

- Implement aggressive retargeting—if user engages with ANY content, they're close to your content in network space
- Build lookalike audiences from seemingly unrelated genres (they're actually connected)
- Create recommendation engines that traverse 2-3 network steps for suggestions
- Use network distance (not just genre match) for programmatic ad targeting

5 Performance Optimization Framework

Campaign Strategy Matrix

Campaign Objective	Target Centrality	KPI Focus
Brand Awareness	High Degree/Closeness	Reach, Impressions
Cross-Audience Reach	High Betweenness	Unique Audiences
Conversion	High Popularity	CPA, ROAS
Content Discovery	High PageRank	Click-Through Rate
Viral Potential	High Clustering	Share Rate

Table 1: Centrality-Based Campaign Targeting Strategy

Budget Allocation Recommendations

Resource Distribution
40% - High-popularity content (proven converters)
30% - High-centrality content (audience expansion)
20% - High-betweenness content (bridge campaigns)
10% - Testing low-centrality niche content (long-tail strategy)

6 Technical Implementation

Data Processing Pipeline

- Data Integration:** Merged 200 popular + 200 top-rated movies, removed 16 duplicates
- Network Construction:** Created 383 nodes with 33,349 edges based on genre overlap
- Centrality Calculation:** Computed 4 metrics using NetworkX (sampled betweenness for speed)
- Normalization:** Min-max normalized all metrics to create composite centrality score
- Correlation Analysis:** Identified metric relationships and divergent patterns

Key Technical Metrics

Network Statistics:	Centrality Ranges:
• Nodes: 383	• Degree: 0.008–0.861
• Edges: 33,349	• Closeness: 0.393–0.878
• Density: 0.456	• Betweenness: 0.000–0.013
• Connected: Yes	• PageRank: 0.001–0.007

7 Actionable Insights for Ana Sofía

Strategic Recommendations

Immediate Actions (Week 1-4):

1. Audit current campaign portfolio: calculate centrality scores for all promoted content
2. Identify high-betweenness content for immediate cross-promotional campaigns
3. Segment audiences by primary genre, but create bridge campaigns targeting genre overlap
4. Update targeting strategy: add "drama" keywords to all campaigns regardless of primary genre

Medium-Term Initiatives (Month 2-6):

1. Build lookalike audiences based on network distance, not just demographic similarity
2. Develop A/B tests comparing centrality-based vs popularity-based content promotion
3. Create "genre journey" campaigns that walk users through connected content (1-3 network steps)
4. Implement dynamic budget allocation: shift spend to high-centrality during awareness phases

Long-Term Strategy (6-12 Months):

1. Partner with content teams to influence production: recommend multi-genre positioning
2. Build proprietary recommendation system using network centrality + collaborative filtering
3. Develop predictive models: forecast content centrality before release based on genre mix
4. Create centrality-based influencer partnerships: match influencers to network position, not just follower count

8 Success Metrics & KPIs

Metric	Baseline	Target (Q2)
Cross-Genre Reach	N/A	+35%
Campaign Efficiency (CPM)	Current	-18%
Content Discovery Rate	12%	22%
Audience Overlap Score	N/A	0.65+
Network-Qualified Leads	N/A	15% of total

Table 2: Network-Strategy Performance Targets

9 Limitations & Future Research

Current Analysis Limitations:

- Genre data is simplified (no sub-genre granularity)
- Temporal dynamics not captured (how networks evolve over time)
- User behavior data not integrated (actual viewing patterns vs theoretical connections)
- Limited to 383 movies (broader catalog would reveal more nuanced patterns)

Next Research Phase:

- Integrate user viewing sequences to validate network predictions
- Add temporal analysis: how does centrality correlate with content lifecycle?
- Expand to 10,000+ movies for statistical robustness
- Include director, actor, and studio networks for deeper analysis

10 Conclusion

Strategic Transformation

Ana Sofia’s network analysis reveals that effective content marketing requires thinking beyond popularity metrics. By understanding how content structurally positions within genre ecosystems, marketers can:

- Predict influence potential before content launches
- Optimize budget allocation based on strategic network position
- Design campaigns that leverage genre bridges for maximum reach
- Build recommendation systems that respect network topology

The key insight: **Genre diversity creates network power.** Multi-genre content isn’t just appealing to more audiences—it structurally occupies more influential positions in the content ecosystem. This transforms content strategy from ”what’s popular?” to ”what connects communities?”

383	33,349	4	0.456
Movies Analyzed	Network Connections	Centrality Metrics	Network Density

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